Native F4/F5, F6 and root solvers

Native F4/F5, F6 and both root-solving families; bounded public Koblitz m2 Boolean fixtures. Final same-host table: run 20261019-024352-boolean-native-policy-ty-fb16d2. Five admitted samples per arm, four primary fixtures, fresh paired planted holdouts231/232. Complete roots and first-witness decomposition have separate boundaries. Every output is checked against complete scalar roots or the original group equation.

Final host: Intel(R) Xeon(R) Platinum 8488C; rustc 1.99.1 (58cf8f9c3 2026-10-13); reserved CPUs 3,7; c7i.2xlarge on-demand, one Rayon worker, exclusive reserved physical core, isolation receipts and A/A. VM neighbours/frequency remain outside the reservation. Failed and contended attempts are retained, excluded, and counted in each JSON summary.

n17: icv1-f2m17-tm101-00378d4e,16 Boolean variables,205 base points, group order 131,174. n23: icv1-f2m23-tm5197-1f85e9e1,22 Boolean variables,2071 base points, group order 8,393,806. Exact constructor output is fixture-identities.json. n19 is a 36-variable implementation-limit refusal, not an intermediate-size timing.

Planted target uses seed230; generator control is the registered generator. The old driver flag random=true does not generate a random target for its single-target invocation. TARGET_LABEL_CORRECTION.md explicitly corrects that earlier label without rewriting frozen inputs or raw records. These cases do not establish an average-case ranking.

Complete-root runs start from the prepared ANF or JSON representation for that API; system construction and JSON encoding are charged when rebuilt in the decomposition wrapper. Cold decomposition charges fresh target-dependent ANF construction, solving, lifting and verification; Semaev arms additionally charge fresh oracle/table preparation once. Common curve/base/context construction is outside every solver interval. Independent complete-root and final group-audit checks run after elapsed capture for all arms. Instrumented query timings include common driver metadata/wrapper overhead and are not bare-kernel latency. No whole-IC/rho ratio is supplied.

F6 is compared through the repository's first-witness decomposition APIs; no complete-original-root F6 API is benchmarked here. The original n23 planted system has two Boolean roots and the generator system has none. Exact complete-root output is a separate obligation from returning one group witness.

The original single-write matrix route qualifies a small native improvement on the primary fixture. Extra SIMD table construction is neutral at both measured sizes and remains experimental. Semaev batch construction preserves every key, first witness, insertion order and abstract group charge; the n23 cold improvement is about 3.6x. Each before/after ratio is from its own same-host experiment; ratios across hosts are never multiplied or pooled.

Iteration1 generalized JSON improvements borrow input, simplify exact evaluation and integrate the existing SIMD Gray kernel. Same-candidate ablations found SIMD slower at 16 variables and faster at 22; auto now keeps scalar below 22. Fresh confirmation measures that policy and the typed F2BoolPoly entry, which avoids JSON/BigUint conversion while retaining exact counts, output caps, budgets, original coordinates and root verification. This is engineering integration, not a new enumeration algorithm.

Single-write, SIMD tables, serial row clearing and batched Semaev setup remain opt-in; extra SIMD tables were neutral. The first serial guard regressed one four-worker control; the refined guard requires explicit RAYON_NUM_THREADS=1 and a current one-worker pool. Generalized auto uses scalar below22 variables and wide Gray when available at22 or more. Typed input avoids JSON; exact counts and original-root verification remain enforced. The18-arm costs precede serial-clear experiments and do not claim their complete-root speedup.

Fresh confirmation retains all outcomes: all 16 typed cells qualify faster than iteration1; n17 prepared JSON roots qualify faster on all four primary/holdout cases. n23 prepared JSON controls are neutral. One n23 planted JSON decomposition control regresses about 3.5% (before/after 0.9654); it is not hidden or described as regression-free. The typed route qualifies about 6x against that frozen wrapper on the same fixture.

Native comparison/admission, statistics and ratios are computed by the Rust analyzers. Five paired ratios use all 3125 bootstrap resamplings with percentile indices78/3046; qualification additionally requires effect greater than max(2%, twice median absolute A/A noise). Charts show recorded medians and min/max; full CI, individual rounds and rejected attempts remain in JSON/raw archive. Python only renders these prepared values.

Validation:13 standalone generalized-solver tests, strict Clippy, example build/tests, exact batched-map and binary/prime backend tests. Before the merge, 4,398 nonignored library tests were covered by passing executions: the8-thread suite had one unchanged time-budget failure, which passed serially; a separate serial full run passed4397 before the new map test. Post-merge latest-candidate serial run:4,397 passed,10 failed,118 ignored; nine failures were sandbox-denied network binds and one was an unchanged wall-threshold test. The network suite then passes38/38 with network access and the exact wall test passes serially. All 4,407 nonignored tests are covered by passing executions. The refined explicit-one-worker guard then passes a complete network-enabled serial library run:4,407 passed,0 failed,118 ignored, exit0. Its four worker/switch kernel configurations and SDK/Clippy/example checks pass. Earlier failing and interrupted invocations remain retained with their actual outcomes.

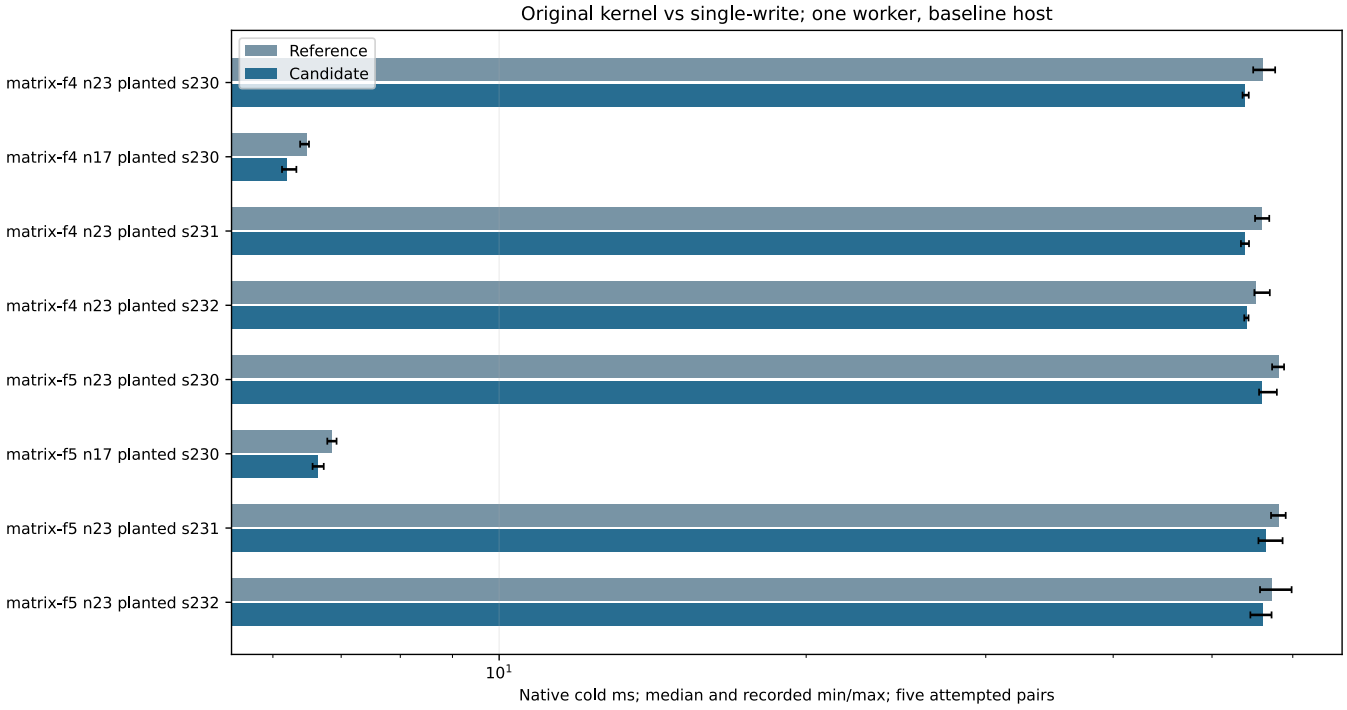
Limits: no external Magma/msolve/FGb/Singular/PolyBoRi runtime comparison, Arm64/GPU measurement or whole-IC/rho promotion. The generalized native field/domain and full eight-coordinate S3 separator are functionally implemented; these m2 timings do not qualify m5 separator performance or the old20-case/8x hard all-roots research gate. The open problem record remains authoritative.

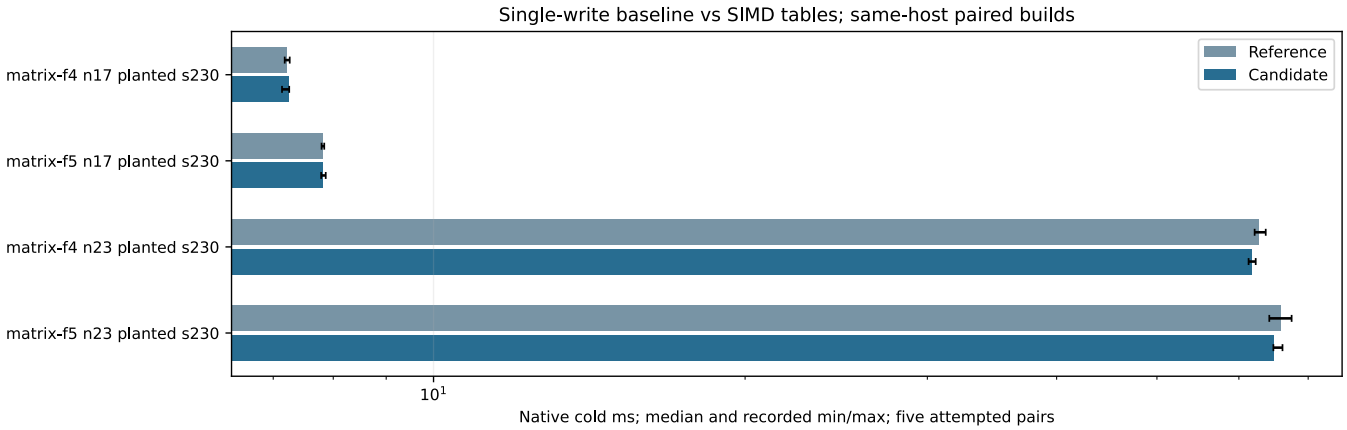
Next measured bottlenecks: repeated F4/F5 matrix construction and recursive complete-root solving, plus the remaining native search kernel in typed generalized decomposition. The fresh typed wrapper is close to its prepared-input root cost; conversion/normalization was the dominant avoidable wrapper cost. Preserve output/accounting contracts and qualify any further layout or construction change with the same paired controls. Leaderboard component panels and browser are regenerated; tables A-C, curve facts and the IC/rho progress series are unchanged because the unit/boundary differs.

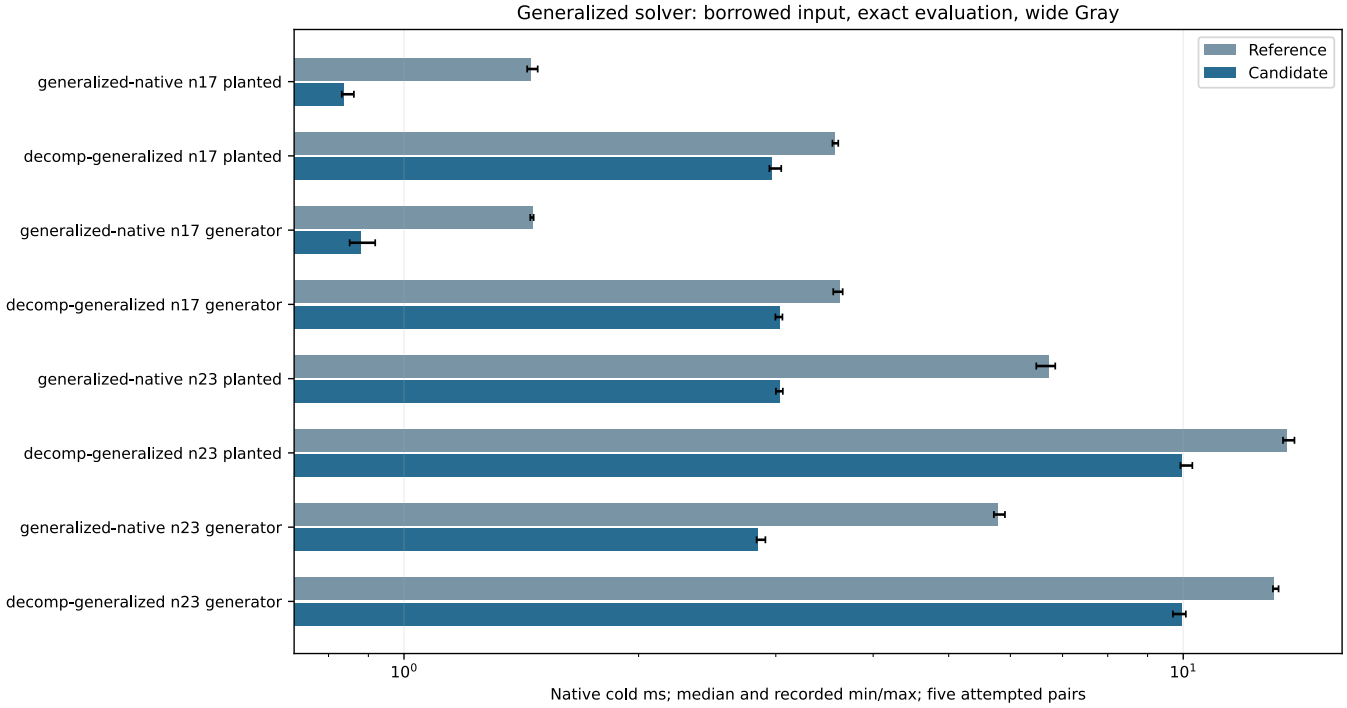
Evidence: initial-summary.json and PROTOCOL.md (run 014036 baseline); optimization-summary.json and OPTIMIZATION_PROTOCOL.md (run 013858 first candidate); final-summary.json and FINAL_PROTOCOL.md (run 014050 Semaev); policy-summary.json, policy-comparison-summary.json and POLICY_PROTOCOL.md (run 024352 confirmation). Source/binary hashes, protocols, commands, every retry, preparation failures and corrected/failed auxiliary Callgrind profiles are retained. Callgrind Ir is not native runtime. See RAW_EVIDENCE.md for archive extraction and immutable job locations.

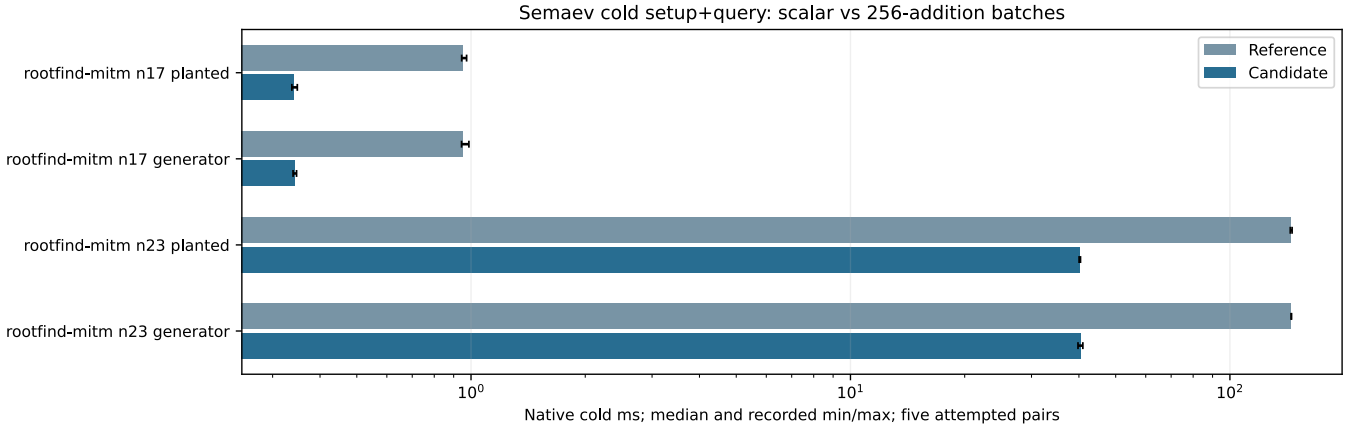
The initial serial-clear matrix trial qualifies all eight one-worker cells faster but regresses the n17 F4 four-worker control (ratio0.8436). The unrestricted registry-query guard is not regression-free. serial-clear-summary.json preserves all outcomes. The refined source requires explicit one-worker configuration before querying Rayon; its fresh confirmation and direct legacy ablation have a separate protocol, summary and report. The eighteen-arm solver costs above precede both experiments with serial clearing off; they are not silently repriced.

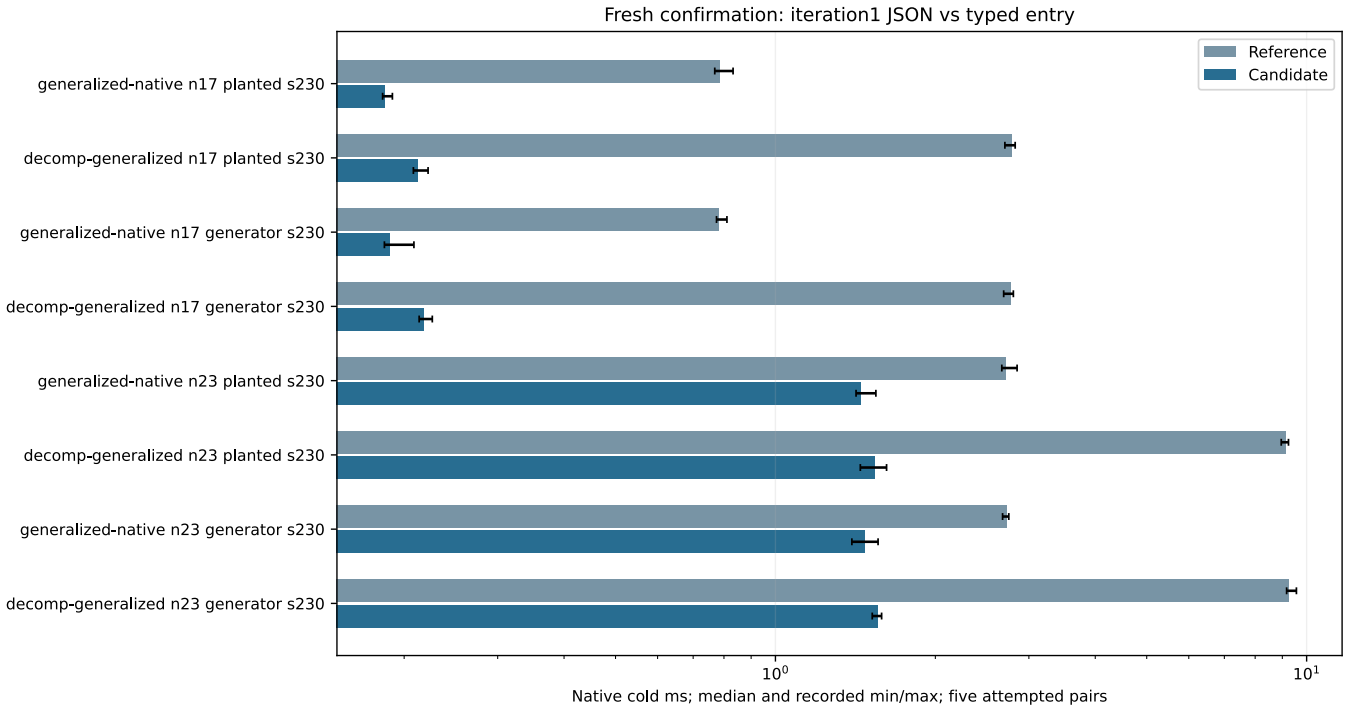
The fresh explicit-one-worker confirmation qualifies all eight one-worker incremental cells faster; all four four-worker controls are neutral. Primary n23 incremental ratios are1.0802 F4 and1.0730 F5. Direct legacy to candidate ratios on this same host are1.1141 F4 and1.1209 F5, rather than products of earlier experiments. All12cells have five eligible pairs; eleven rejected attempts remain. Activation stays opt-in. These matrix-call gains do not establish a complete-root or decomposition improvement.

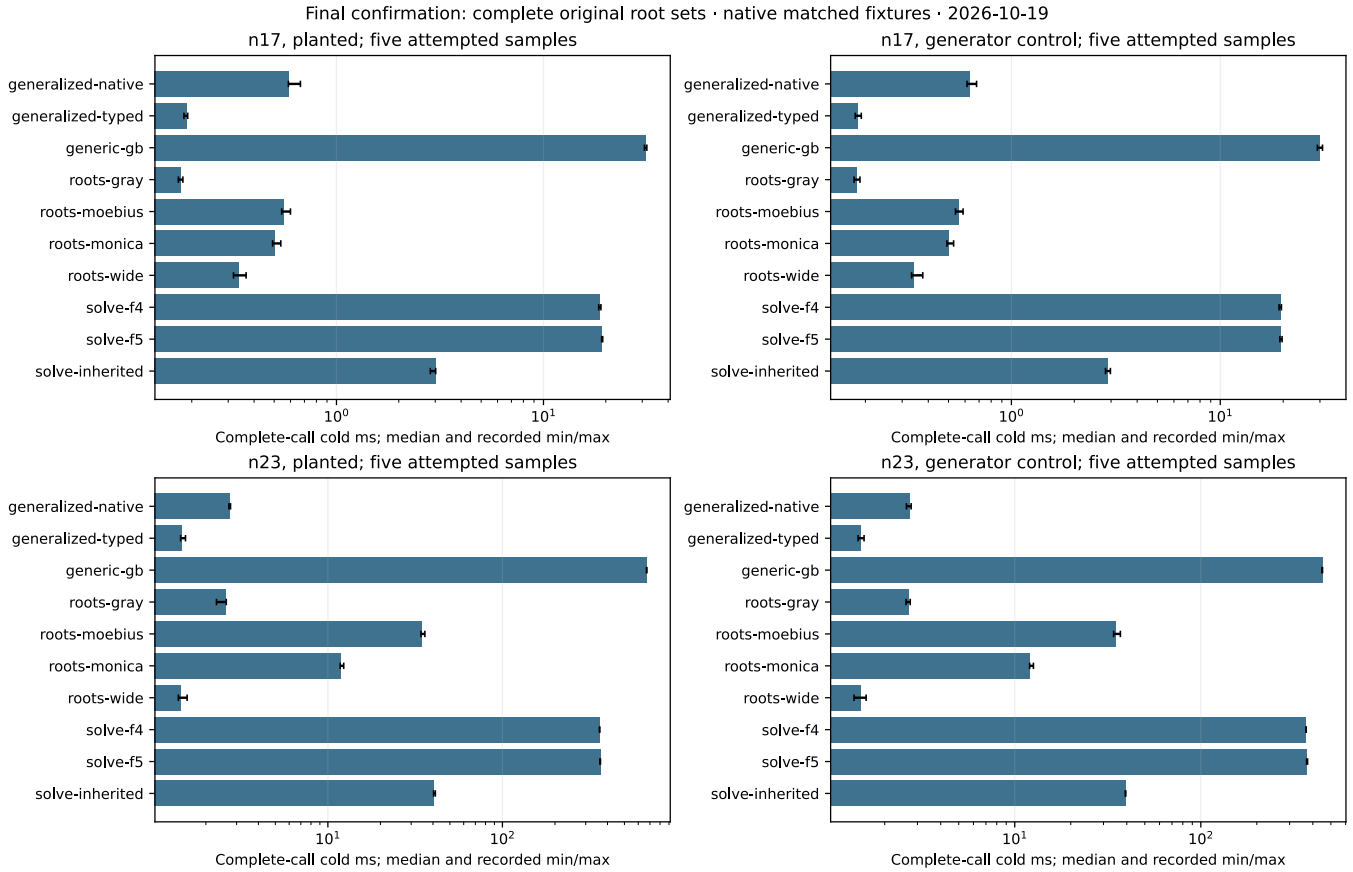


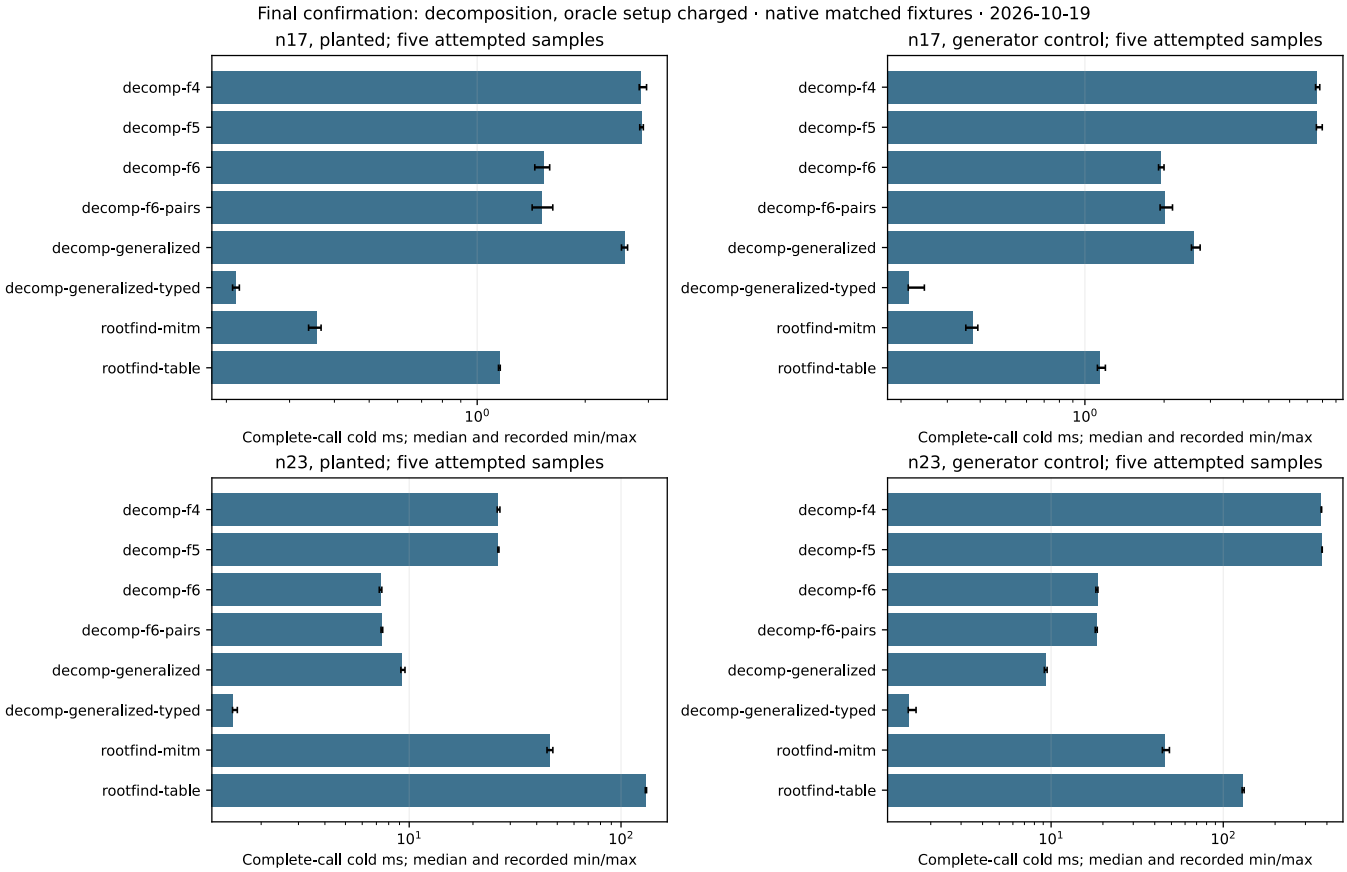

Single-write baseline vs SIMD tables; same-host paired builds
Reference
Candidate
matrix-f4 n17 planted s230
matrix-f5 n17 planted s230
matrix-f4 n23 planted s230
matrix-f5 n23 planted s230
10^1
Native cold ms; median and recorded min/max; five attempted pairs

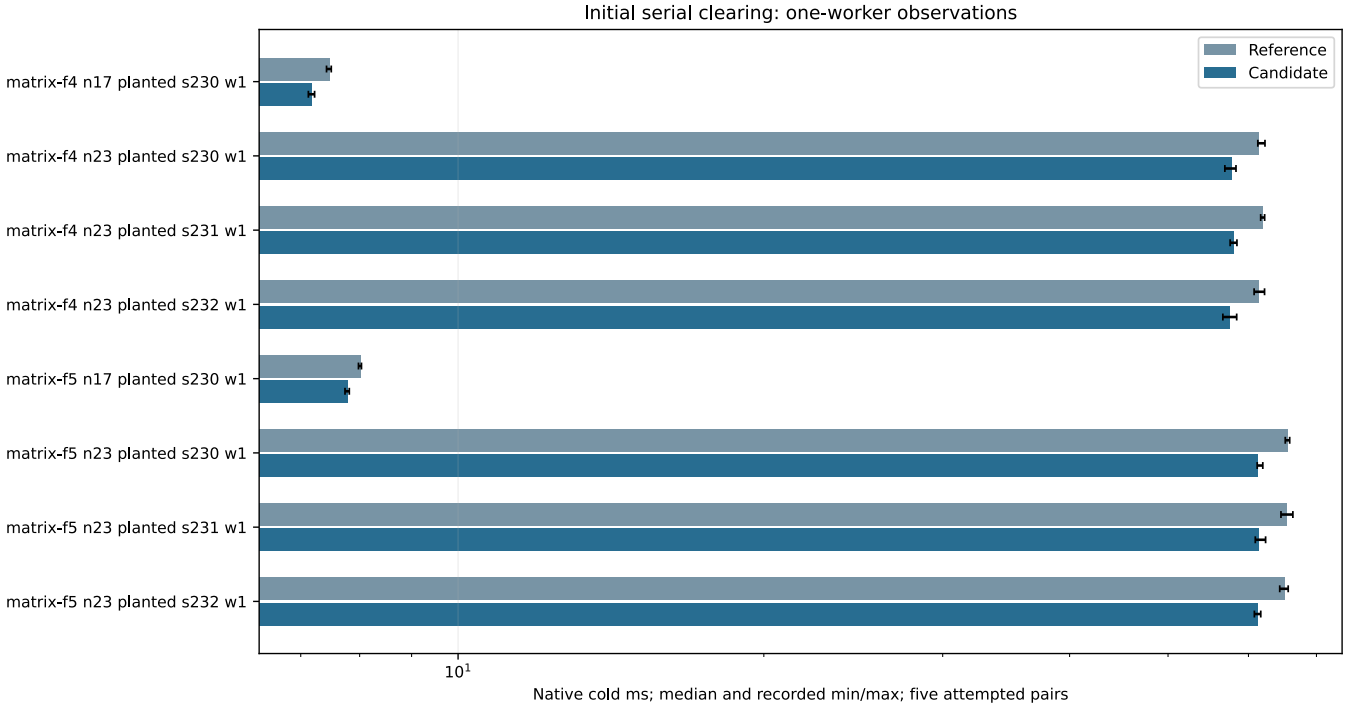

Generalized solver: borrowed input, exact evaluation, wide Gray
Reference
Candidate
generalized-native n17 planted
decomp-generalized n17 planted
generalized-native n17 generator
decomp-generalized n17 generator
generalized-native n23 planted
decomp-generalized n23 planted
generalized-native n23 generator
decomp-generalized n23 generator
10^0
10^1
Native cold ms; median and recorded min/max; five attempted pairs

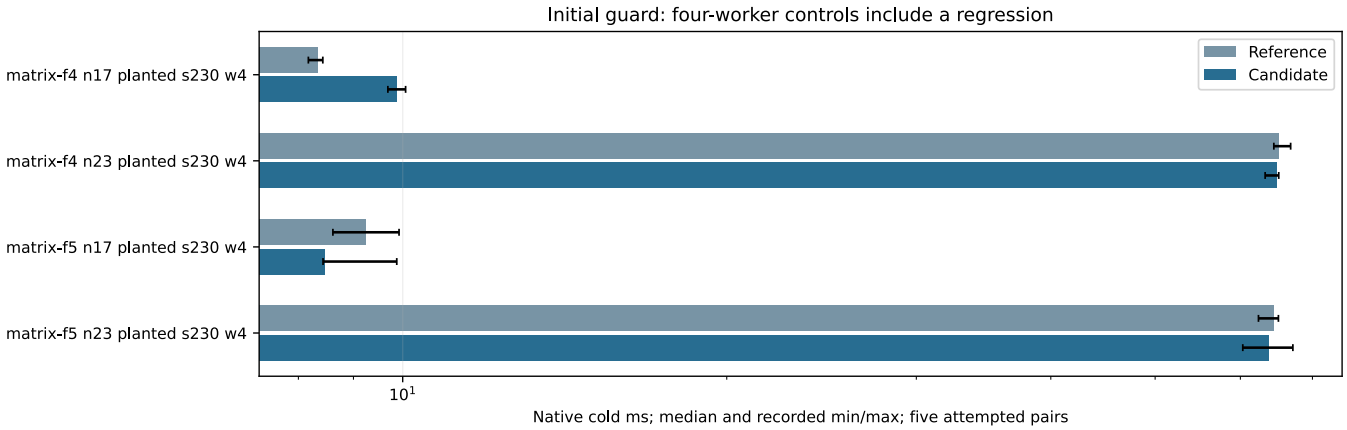

Semaev cold setup+query: scalar vs 256-addition batches
Reference
Candidate
rootfind-mitm n17 planted
rootfind-mitm n17 generator
rootfind-mitm n23 planted
rootfind-mitm n23 generator
10^0
10^1
10^2
Native cold ms; median and recorded min/max; five attempted pairs

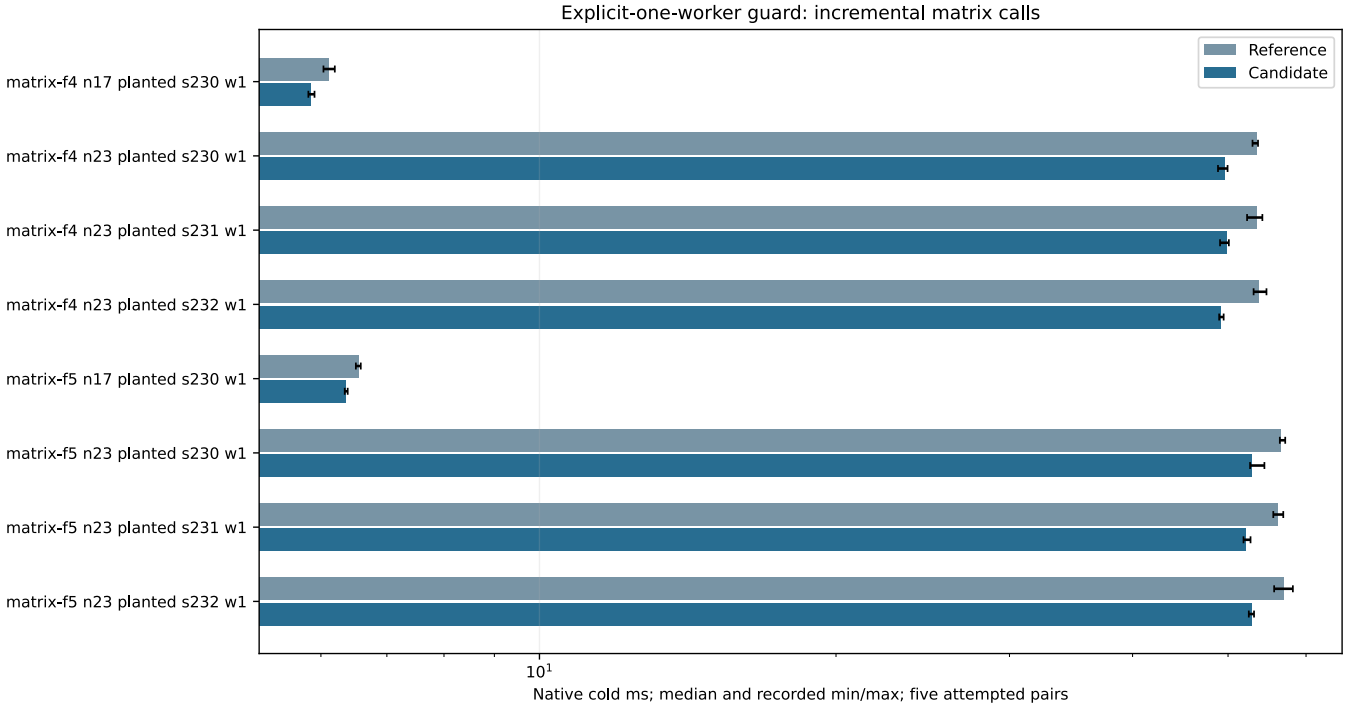

Fresh confirmation: iteration1 JSON vs typed entry
Reference
Candidate
generalized-native n17 planted s230
decomp-generalized n17 planted s230
generalized-native n17 generator s230
decomp-generalized n17 generator s230
generalized-native n23 planted s230
decomp-generalized n23 planted s230
generalized-native n23 generator s230
decomp-generalized n23 generator s230
10^0
10^1
Native cold ms; median and recorded min/max; five attempted pairs

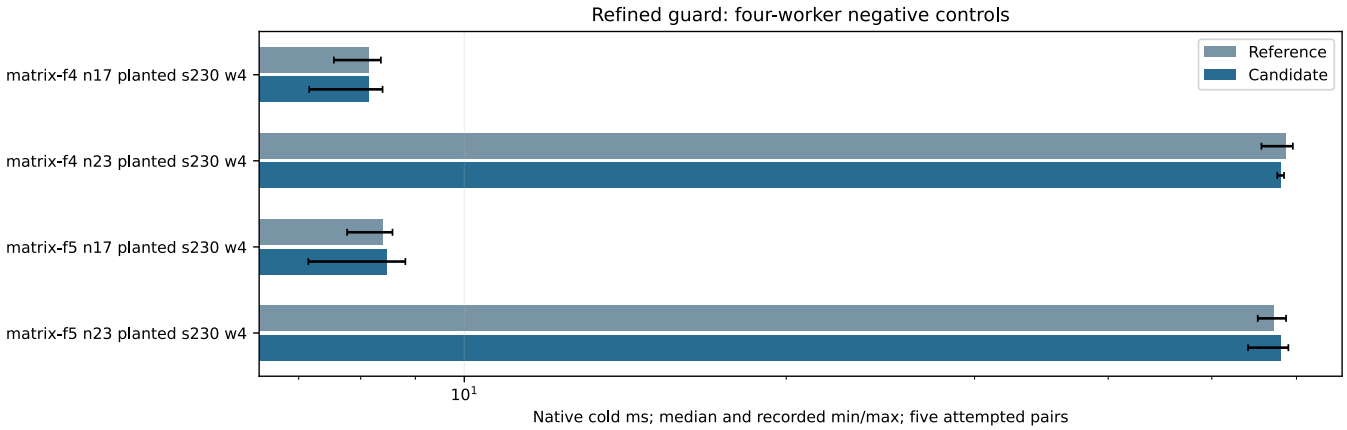

Final confirmation: complete original root sets · native matched fixtures · 2026-10-19
n17, planted; five attempted samples
n17, generator control; five attempted samples
n23, planted; five attempted samples
n23, generator control; five attempted samples
generalized-native
generalized-typed
generic-gb
roots-gray
roots-moebius
roots-monica
roots-wide
solve-f4
solve-f5
solve-inherited
Complete-call cold ms; median and recorded min/max

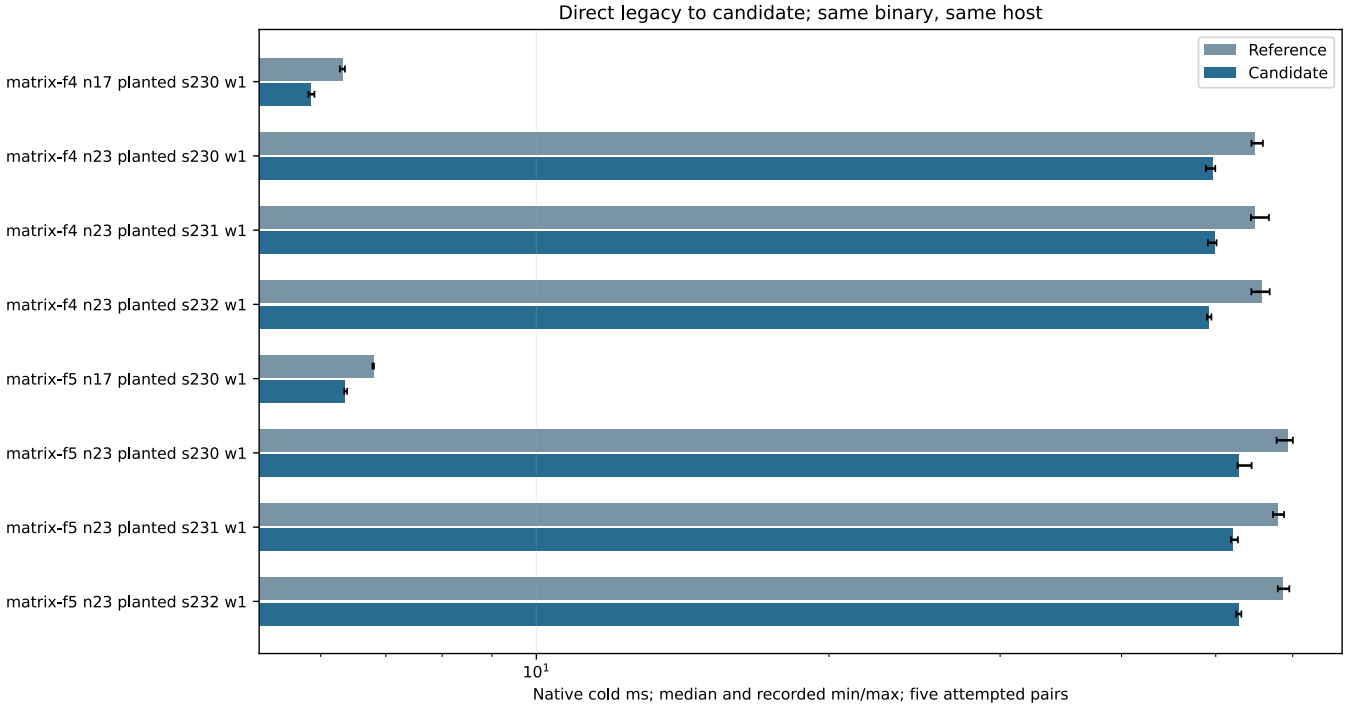

Final confirmation: decomposition, oracle setup charged · native matched fixtures · 2026-10-19
n17, planted; five attempted samples
n17, generator control; five attempted samples
n23, planted; five attempted samples
n23, generator control; five attempted samples
decomp-f4
decomp-f5
decomp-f6
decomp-f6-pairs
decomp-generalized
decomp-generalized-typed
rootfind-mitm
rootfind-table
Complete-call cold ms; median and recorded min/max


Initial serial clearing: one-worker observations
Reference
Candidate
matrix-f4 n17 planted s230 w1
matrix-f4 n23 planted s230 w1
matrix-f4 n23 planted s231 w1
matrix-f4 n23 planted s232 w1
matrix-f5 n17 planted s230 w1
matrix-f5 n23 planted s230 w1
matrix-f5 n23 planted s231 w1
matrix-f5 n23 planted s232 w1
10^1
Native cold ms; median and recorded min/max; five attempted pairs


Initial guard: four-worker controls include a regression
Reference
Candidate
matrix-f4 n17 planted s230 w4
matrix-f4 n23 planted s230 w4
matrix-f5 n17 planted s230 w4
matrix-f5 n23 planted s230 w4
10^1
Native cold ms; median and recorded min/max; five attempted pairs


Explicit-one-worker guard: incremental matrix calls
Reference
Candidate
matrix-f4 n17 planted s230 w1
matrix-f4 n23 planted s230 w1
matrix-f4 n23 planted s231 w1
matrix-f4 n23 planted s232 w1
matrix-f5 n17 planted s230 w1
matrix-f5 n23 planted s230 w1
matrix-f5 n23 planted s231 w1
matrix-f5 n23 planted s232 w1
10^1
Native cold ms; median and recorded min/max; five attempted pairs


Refined guard: four-worker negative controls
Reference
Candidate
matrix-f4 n17 planted s230 w4
matrix-f4 n23 planted s230 w4
matrix-f5 n17 planted s230 w4
matrix-f5 n23 planted s230 w4
10^1
Native cold ms; median and recorded min/max; five attempted pairs


Direct legacy to candidate; same binary, same host
Reference
Candidate
matrix-f4 n17 planted s230 w1
matrix-f4 n23 planted s230 w1
matrix-f4 n23 planted s231 w1
matrix-f4 n23 planted s232 w1
matrix-f5 n17 planted s230 w1
matrix-f5 n23 planted s230 w1
matrix-f5 n23 planted s231 w1
matrix-f5 n23 planted s232 w1
10^1
Native cold ms; median and recorded min/max; five attempted pairs

n23 complete_roots, planted; five samples, ratio relative to F4

Method	Cold ms	n	F4 / method
generalized-native	2.7533	5	131.150
generalized-typed	1.4494	5	248.149
generic-gb	669.8408	5	0.538
roots-gray	2.5987	5	138.912
roots-moebius	34.4747	5	10.448
roots-monica	11.9093	5	30.193
roots-wide	1.4275	5	252.955
solve-f4	360.1913	5	1.000
solve-f5	363.9447	5	0.991
solve-inherited	40.4737	5	8.896

n23 complete_roots, generator; five samples, ratio relative to F4

Method	Cold ms	n	F4 / method
generalized-native	2.7096	5	135.873
generalized-typed	1.4774	5	249.159
generic-gb	449.6532	5	0.818
roots-gray	2.7031	5	136.175
roots-moebius	34.8805	5	10.553
roots-monica	12.0878	5	30.491
roots-wide	1.4900	5	247.049
solve-f4	368.1564	5	1.000
solve-f5	372.7797	5	0.988
solve-inherited	39.3968	5	9.342

n23 decomposition, planted; five samples, ratio relative to F4

Method	Cold ms	n	F4 / method
decomp-f4	26.3691	5	1.000
decomp-f5	26.2727	5	1.004
decomp-f6	7.3539	5	3.610
decomp-f6-pairs	7.3983	5	3.552
decomp-generalized	9.2244	5	2.859
decomp-generalized-typed	1.4781	5	17.840
rootfind-mitm	46.3986	5	0.564
rootfind-table	131.0207	5	0.201

n23 decomposition, generator; five samples, ratio relative to F4

Method	Cold ms	n	F4 / method
decomp-f4	370.4942	5	1.000
decomp-f5	375.1102	5	0.988
decomp-f6	18.6067	5	19.910
decomp-f6-pairs	18.3590	5	20.178
decomp-generalized	9.3417	5	39.691
decomp-generalized-typed	1.4931	5	248.331
rootfind-mitm	45.6419	5	8.117
rootfind-table	130.1315	5	2.853

Same-host paired changes; different rows are not pooled

Change	Target	Before ms	After ms	Ratio	Decision
matrix-f4 single-write	planted	56.0625	53.9084	1.0400	faster
matrix-f5 single-write	planted	58.1545	55.9272	1.0363	faster
Semaev batched cold setup	planted	144.6895	40.1599	3.6050	faster
Semaev batched cold setup	generator	145.1419	40.4301	3.5899	faster
n17 generalized typed entry	planted	2.7787	0.2124	12.9877	faster
n17 generalized typed entry	generator	2.7740	0.2178	12.6796	faster
n23 generalized typed entry	planted	9.1164	1.5370	6.0022	faster
n23 generalized typed entry	generator	9.2590	1.5568	6.0379	faster
initial n17 F4 four-worker control	planted	8.3357	9.8786	0.8436	slower
matrix-f4 guarded serial clearing	planted	53.4171	49.6467	1.0802	faster
matrix-f4 legacy to guarded candidate	planted	54.9114	49.6467	1.1141	faster
matrix-f5 guarded serial clearing	planted	56.6003	52.8025	1.0730	faster
matrix-f5 legacy to guarded candidate	planted	59.2979	52.8025	1.1209	faster

Fresh cutoff and typed-entry confirmation, all holdouts

Arm	n	Seed	Target	Old/JSON	Decision	Old/typed	Decision
generalized-native	17	230	planted	1.2909	faster	4.2625	faster
decomp-generalized	17	230	planted	1.0847	faster	12.9877	faster
generalized-native	17	230	generator	1.2461	faster	4.1444	faster
decomp-generalized	17	230	generator	1.0613	neutral	12.6796	faster
generalized-native	17	231	planted	1.2162	faster	4.2937	faster
decomp-generalized	17	231	planted	1.0607	faster	11.5310	faster
generalized-native	17	232	planted	1.2148	faster	4.1460	faster
decomp-generalized	17	232	planted	1.0568	neutral	11.9363	faster
generalized-native	23	230	planted	0.9744	neutral	1.8596	faster
decomp-generalized	23	230	planted	0.9654	slower	6.0022	faster
generalized-native	23	230	generator	1.0099	neutral	1.8610	faster
decomp-generalized	23	230	generator	1.0151	neutral	6.0379	faster
generalized-native	23	231	planted	1.0243	neutral	1.9342	faster
decomp-generalized	23	231	planted	0.9830	neutral	6.0849	faster
generalized-native	23	232	planted	0.9985	neutral	1.9497	faster
decomp-generalized	23	232	planted	1.0136	neutral	6.0081	faster